



Flappy Bird Game

p5play JS library



Lesson 7 Display Score

4 Sept 2025

Why 2-3 lessons

- Lots of concepts to teach
- Arrays to manage a list of digit images
- Check each digit image size: what is the width and height
- Conversion from number to String, to Array
- Loop through an Array

Step 1: Use JS Array instead of using 10 variables, to load 10 digits

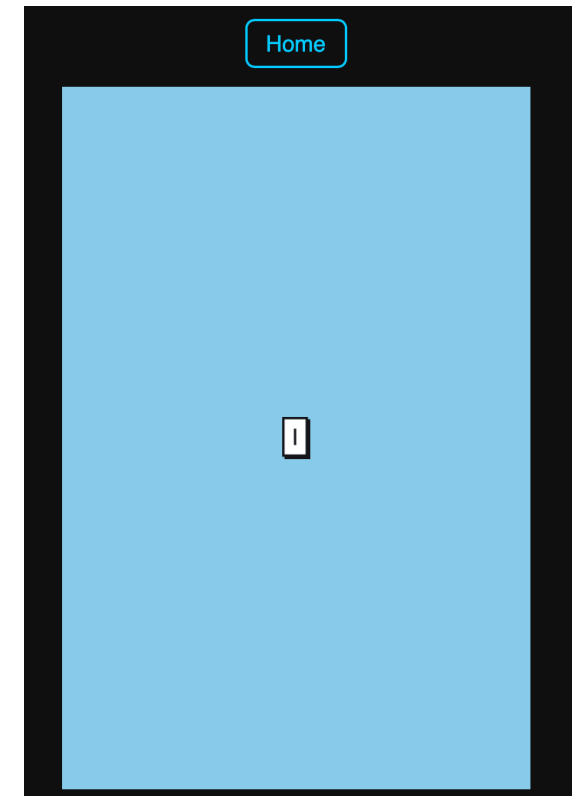
```
1.let digitImgs = [];  
2.let digit0;  
  
4.let scoreNumber = 0;  
5.let scoreSprite;  
6.
```

> In the preload() function, load a single digit

```
10.  digit0 = loadImage('assets/0.png');
```

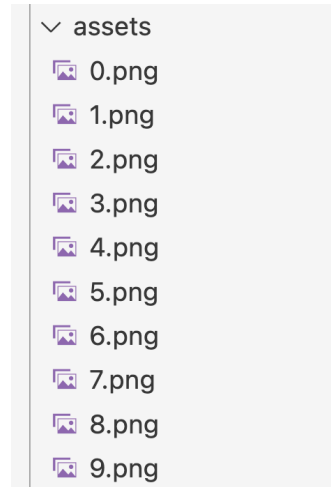
> Use a sprite to load the 0 digit image

```
30.function setup() {  
31.  new Canvas(400,600);  
  
33.  scoreSprite = new Sprite();  
34.  scoreSprite.img = digit0;  
35.  scoreSprite.collider = "none"; // explain static, dynamic, none  
36.}  
  
38.function draw() {  
39.  background("skyblue"); // erase the trail left behind  
40.}  
41.
```



Step 2: Use 10 variables, to load 10 digits

```
1.function preload() {  
2.    digit0 = loadImage('assets/0.png');  
3.    digit1 = loadImage('assets/1.png');  
4.    digit2 = loadImage('assets/2.png');  
5.    digit3 = loadImage('assets/3.png');  
6.    digit4 = loadImage('assets/4.png');  
7.    digit5 = loadImage('assets/5.png');  
8.    digit6 = loadImage('assets/6.png');  
9.    digit7 = loadImage('assets/7.png');  
10.   digit8 = loadImage('assets/8.png');  
11.   digit9 = loadImage('assets/9.png');
```



***Take note of the filenames.**

Step 2: Use 10 variables, to load 10 digits

```
1.function preload() {
2.    digit0 = loadImage('assets/0.png');
3.    digit1 = loadImage('assets/1.png');
4.    digit2 = loadImage('assets/2.png');
5.    digit3 = loadImage('assets/3.png');
6.    digit4 = loadImage('assets/4.png');
7.    digit5 = loadImage('assets/5.png');
8.    digit6 = loadImage('assets/6.png');
9.    digit7 = loadImage('assets/7.png');
10.   digit8 = loadImage('assets/8.png');
11.   digit9 = loadImage('assets/9.png');

13.   let prefix = 'assets/';
14.   let postfix = '.png';
15.   let filename;
16.   for (let count=0; count<10; count++) {
17.       filename = prefix + count + postfix;
18.       console.log(filename);
19.   }
20.}
21.
```

Check JavaScript console in the Web Browser

Console opened at 11:19:06	
assets/0.png	preload — scores.js:26
assets/1.png	preload — scores.js:26
assets/2.png	preload — scores.js:26
assets/3.png	preload — scores.js:26
assets/4.png	preload — scores.js:26
assets/5.png	preload — scores.js:26
assets/6.png	preload — scores.js:26
assets/7.png	preload — scores.js:26
assets/8.png	preload — scores.js:26
assets/9.png	preload — scores.js:26

Not efficient if I have to use 10 variables for 10 digits.

Is there a better way?



Step 3: Change to use a JS Array

```
1.// first lesson to use arrays
2.// go slow
3.let digitImgs = [];
4.let digit0;
```

Empty JS array
Use []

```
6.let scoreNumber = 0;
7.let scoreSprite;
```

```
9.function preload() {
10.    // load 10 digits into a JS Array
11.    let prefix = 'assets/';
12.    let postfix = '.png';
13.    let filename;
14.    for (let count=0; count<10; count++) {
15.        filename = prefix + count + postfix;
16.        // console.log(filename);
17.        digitImgs[count] = loadImage(filename);
18.    }
19.}
```

Each image is loaded one by one into
the array using the index number

```
21.function setup() {
22.    new Canvas(400,600);
```

```
24.    scoreSprite = new Sprite();
25.    scoreSprite.img = digitImgs[0];
26.    scoreSprite.collider = "none"; // explain static, dynamic, none
27.}
28.
```

Use the first one in the array.

Step 4: Add key pressed to change the score value

```
29.function draw() {  
30.    background("skyblue"); // erase the trail left behind  
  
32.    if (kb.presses("up")) {  
33.        scoreNumber++;  
34.    }  
  
36.    if (kb.presses("down")) {  
37.        scoreNumber--;  
38.    }  
  
40.    // keep the variable within 0-9  
41.    scoreNumber = constrain(scoreNumber, 0, 9);  
  
43.    textSize(16);  
44.    text("Score is " + scoreNumber, 50,50);  
45.}  
46.
```



Checking for up arrow key pressed



constrain() function can restrict the values of any variable.

This is for testing purposes.

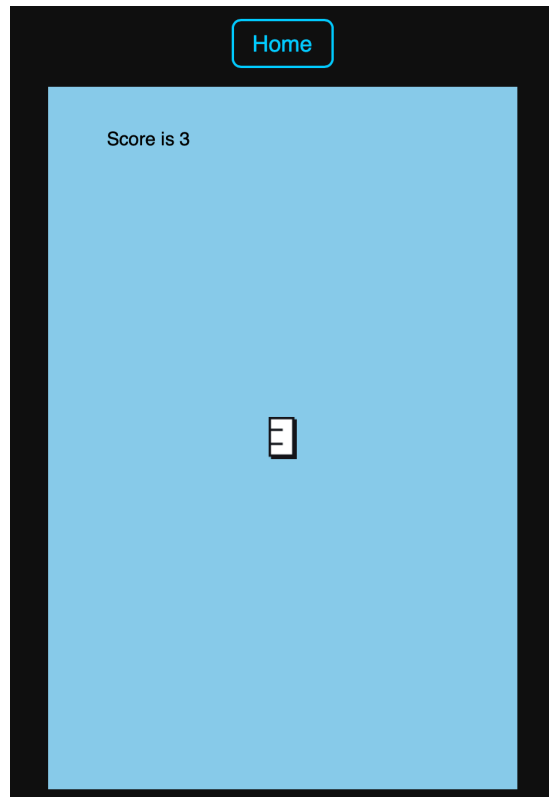
We want to see it on the screen so that we can change our digits images too.

Step 5: as the variable value changes, change the digit images

```
40. // keep the variable within 0-9
41. scoreNumber = constrain(scoreNumber, 0, 9);
42. scoreSprite.img = digitImgs[scoreNumber];
43.
```



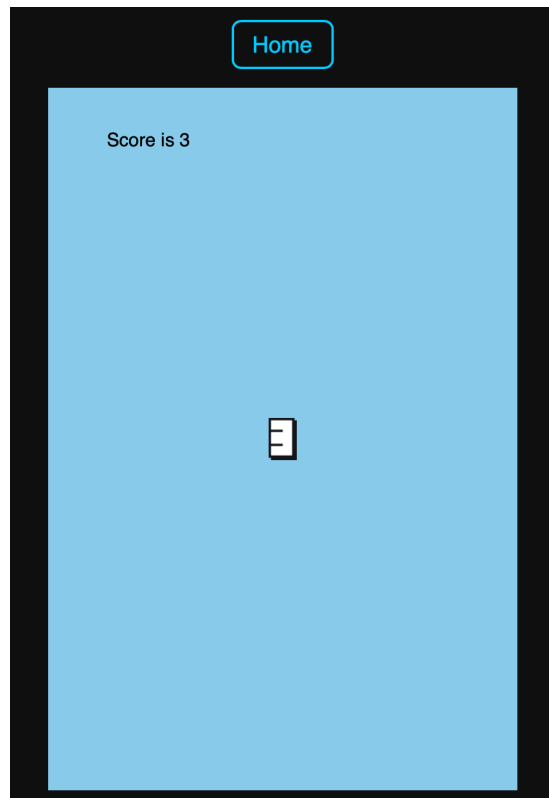
Set the correct digit image



Press up or down arrow keys
Observe how the score variable
and digits images are updating too.

Step 5: as the variable value changes, change the digit images

```
40. // keep the variable within 0-9
41. scoreNumber = constrain(scoreNumber, 0, 9);
42. scoreSprite.img = digitImgs[scoreNumber];
43.
```



Press **up** or **down** arrow keys
Observe how the score variable
and digits images are updating too.

What if my score goes
beyond 9?



Step 6: Handle 2 digits or even 3 digits score (part 1)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array



Index position: 0 1

p5 JS library has convenient methods

p5.js

Reference
Tutorials
Examples
Contribute
Community
About

English

Accessibility

Reference > constrain()

constrain()

Constrains a number between a minimum and maximum value.

p5.js

Reference
Tutorials
Examples
Contribute
Community
About

English

Accessibility



Reference > str()

str()

Converts a Boolean or Number to String.

str() converts values to strings. See the String reference page for guidance on using template literals instead.

The parameter, n, is the value to convert. If n is a Boolean, as in str(false) or str(true), then the value will be returned as a string, as in 'false' or 'true'. If n is a number, as in str(123), then its value will be returned as a string, as in '123'. If an array is passed, as in str([12.34, 56.78]), then an array of strings will be returned.

</> Start Coding

Donate

Reference

Step 6: Handle 2 digits or even 3 digits score (part 2)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array

```
42. // keep the variable within 0-9
43. scoreNumber = constrain(scoreNumber, 0, 999);

45. // call custom function
46. drawScore();
```

New custom function

To make it easy to merge codes later into flappybird.js file.

```
53. // use function to group codes within a context together
54. function drawScore() {
55.   // scoreSprite.img = digitImgs[scoreNumber];
56.   let scoreString = str(scoreNumber);
57.   // console.log("score as a string: " + scoreString);
58.   let scoreDigitArray = scoreString.split("");
59.
60.   text("Score array size is " + scoreDigitArray.length, 50, 70);
61.   text("Score array at index 1 is " + scoreDigitArray[0], 50, 90);
63. }
64.
```

4	8
---	---

Index position: 0 1

Debugging the variables

JavaScript has convenient methods too

In JavaScript, several methods can be used to convert a string into an array, depending on the desired outcome: Using `split()`.

This is the most common and versatile method for splitting a string into an array of substrings based on a specified `separator`.

JavaScript



```
const myString = "apple,banana,orange";
const myArray = myString.split(","); // Splits by comma
console.log(myArray); // Output: ["apple", "banana", "orange"]

const anotherString = "hello world";
const charArray = anotherString.split(""); // Splits by each character
console.log(charArray); // Output: ["h", "e", "l", "l", "o", " ", "w", "o", "r", "l", "d"]
```

Step 6: Handle 2 digits or even 3 digits score (part 3)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array



Index position: 0 1

```
42.  // keep the variable within 0-9
43.  scoreNumber = constrain(scoreNumber, 0, 999);

45.  // call custom function
46.  drawScore();
```

```
53. // use function to group codes within a context together
54. function drawScore() {
55.   // scoreSprite.img = digitImgs[scoreNumber];
56.   let scoreString = str(scoreNumber);
57.   // console.log("score as a string: " + scoreString);
58.   let scoreDigitArray = scoreString.split("");
59.
```

```
61.   text("Score array size is " + scoreDigitArray.length, 50,70);
62.   text("Score array at index 1 is " + scoreDigitArray[0], 50,90);

64. }
65.
```

 Split the String into an Array

Step 6: Handle 2 digits or even 3 digits score (part 4)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array

	4	8
Index position:	0	1



How do you get the digit image for index position 0?



Step 6: Handle 2 digits or even 3 digits score (part 5)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array

	4	8
Index position:	0	1



How do you get the digit image for index position 1?



Step 6: Handle 2 digits or even 3 digits score (part 6)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array

4	8
---	---

Index position: 0 1

How many sprites do
you need?



Step 6: Handle 2 digits or even 3 digits score (part 7)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array

	4	8
Index position:	0	1



Step 6: Handle 2 digits or even 3 digits score (part 8)

- Score variable as a number, is easier for counting points in a game
- But for displaying, we can convert it to a String for ease of taking out each digit
- From String, its easy to break it up into JS Array

	4	8
Index position:	0	1

Use the size of the
JS Array

2 Sprites



Index position: 0 1

4	8
---	---

Step 7: Use Sprite Group (part 1)

```
1. let digitImgs = [];
```

```
3. let scoreNumber = 11;
```

```
4. let scoreSprite;
```

```
5. let scoreGroup;
```

```
6.
```

Use Sprite Group

```
21. function setup() {
```

```
22.   new Canvas(400,600);
```

```
24.   // no need
```

```
25.   // scoreSprite = new Sprite();
```

```
26.   // scoreSprite.img = digitImgs[0];
```

```
27.   // scoreSprite.collider = "none"; // explain static, dynamic, none
```

```
29.   // use Sprite Group to manage the digits
```

```
30.   scoreGroup = new Group();
```

```
31.   scoreGroup.collider = "none";
```

```
32.   scoreGroup.layer = 1000;
```

```
33. }
```

```
34.
```

Use Sprite Group

Step 7: Use Sprite Group (Part 2)

```
59. // use function to group codes within a context together
```

```
60. function drawScore() {
```

```
62.     // clear previous score digits sprites
```

```
63.     scoreGroup.removeAll();
```



Remove all members

```
65.     // number to a String value
```

```
66.     let scoreString = str(scoreNumber);
```

```
67.     // String to an Array
```

```
68.     let scoreDigitArray = scoreString.split("");
```

```
70.     let offset = 0;
```

```
71.     let middle = width/2; // center of screen
```

```
73.     for (let one of scoreDigitArray) {
```

```
74.         let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);
```

```
75.         oneSprite.img = digitImgs[one];
```

```
76.         offset = offset + 25; // move right by size of image width + 1px
```

```
77.     }
```

```
78. }
```

```
79.
```



Add member to the Group

X position



Y position



```
74.let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);
```



The sprite is added as a member of the Sprite Group



???

X position

Y position

```
74.let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);
```

The sprite is added as a member of the Sprite Group

???

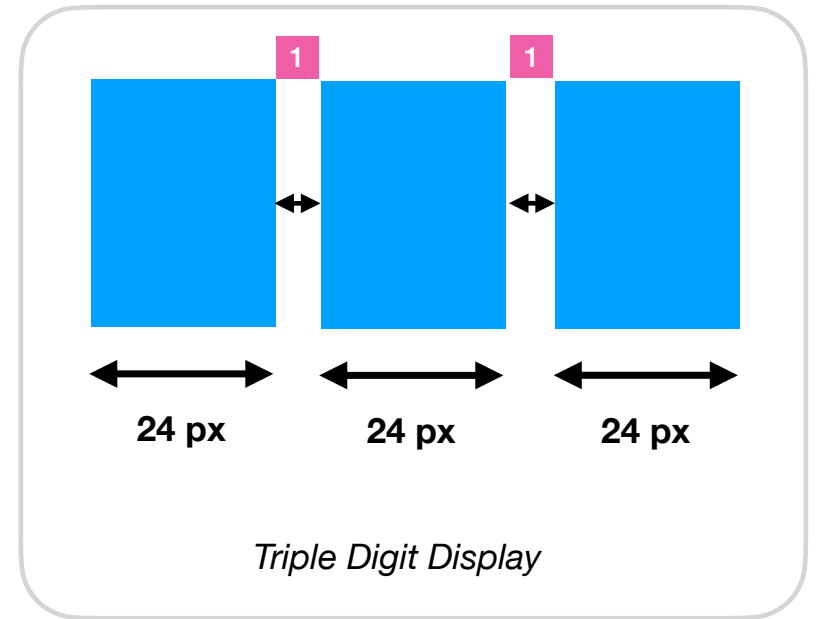
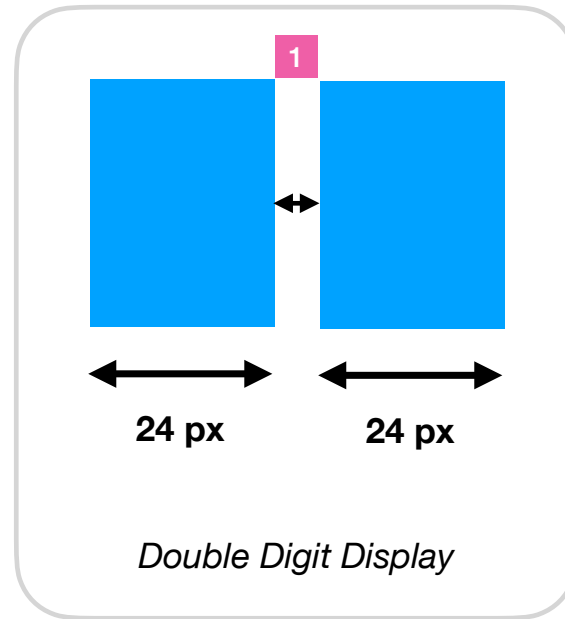
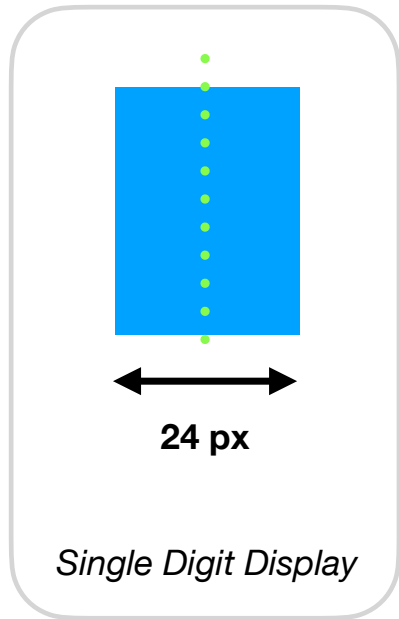
```
70. let offset = 0;
71. let middle = width/2; // center of screen

73. for (let one of scoreDigitArray) {
74.     let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);
75.     oneSprite.img = digitImgs[one];
76.     offset = offset + 25; // move right by size of image width + 1px
77. }
78.
```

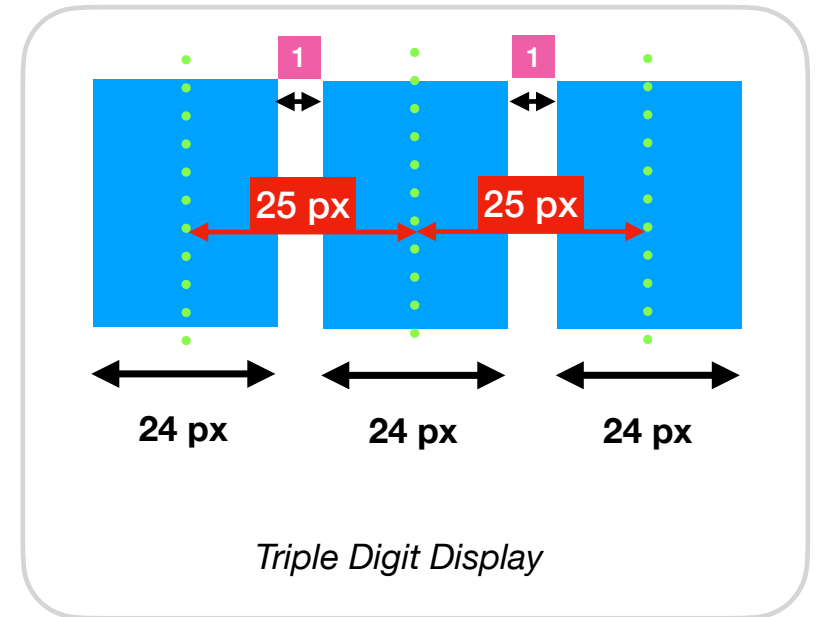
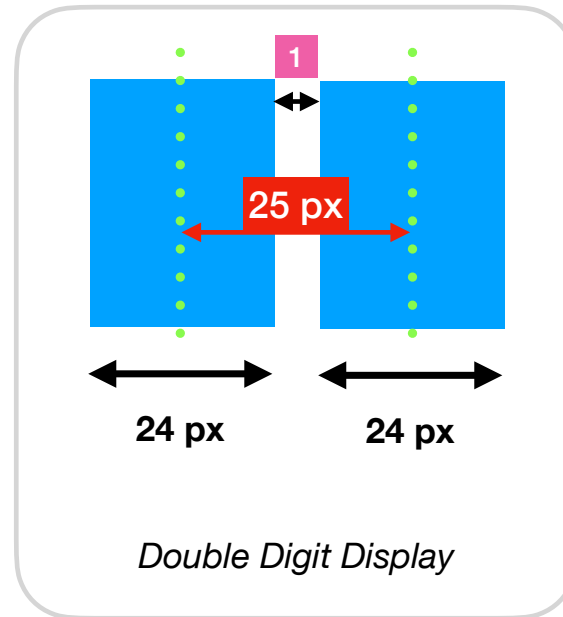
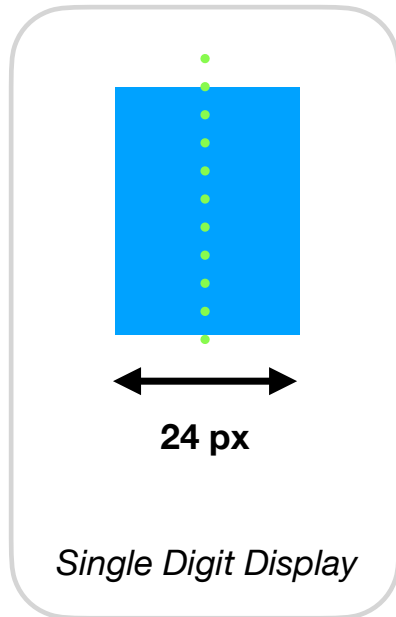


offset is used to add spacing between the digits

1 Leave a gap of 1 px between the digits



Considering the difference of 25 px, we can know each sprite's xpos (centre point)



Any question?



```
70. let offset = 0;
71. let middle = width/2; // center of screen

73. for (let one of scoreDigitArray) {
74.     let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);
75.     oneSprite.img = digitImgs[one];
76.     offset = offset + 25; // move right by size of image width + 1px
77. }
78.
```

Look at line 75
What is the value of "one"?



Task: Add console.log() to print the variable "one".

Step 7: Use Sprite Group (Part 2)

```
59.// use function to group codes within a context together
```

```
60.function drawScore() {
```

```
62.    // clear previous score digits sprites
```

```
63.    scoreGroup.removeAll();
```



Remove all members

```
65.    // number to a String value
```

```
66.    let scoreString = str(scoreNumber);
```

```
67.    // String to an Array
```

```
68.    let scoreDigitArray = scoreString.split("");
```

```
70.    let offset = 0;
```

```
71.    let middle = width/2; // center of screen
```

```
73.    for (let one of scoreDigitArray) {
```

```
74.        let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);
```

```
75.        oneSprite.img = digitImgs[one];
```

```
76.        offset = offset + 25; // move right by size of image width + 1px
```

```
77.    }
```

```
78.}
```

```
79.
```



Add member to the Group

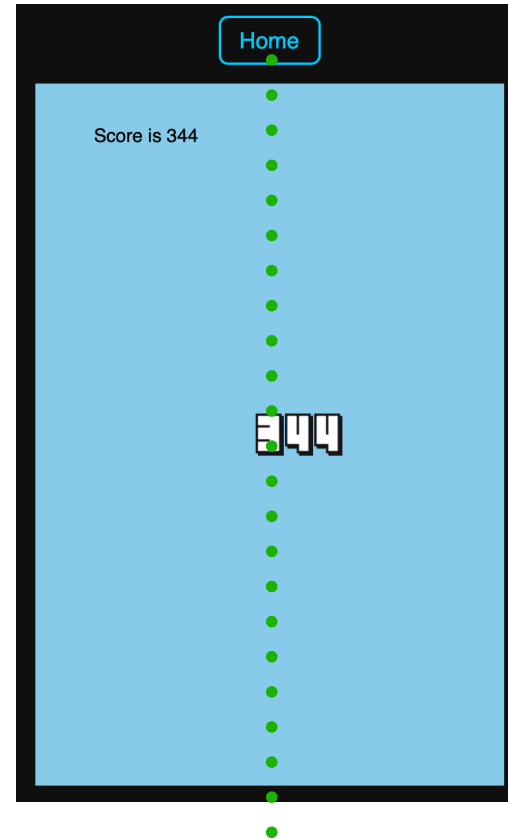
WHAT IF your score to a 3 digits number???

Observe: The digits are not in the centre of the screen.

```
44.  if (kb.presses("3")) {  
45.      scoreNumber = 345;  
46.  }  
47.  // keep the variable within 0-999  
48.  scoreNumber = constrain(scoreNumber, 0, 999);  
49.
```



A quick way to increase the score to 3 digits value.

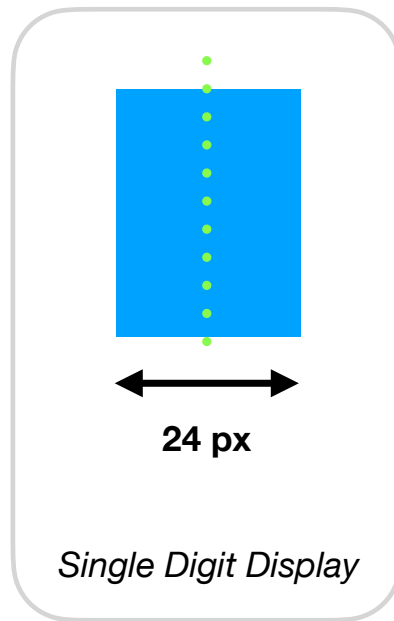


Display Score in the Center

Explanation

Display Score in the Center of the Game

Easy. Just follow `camera.x`

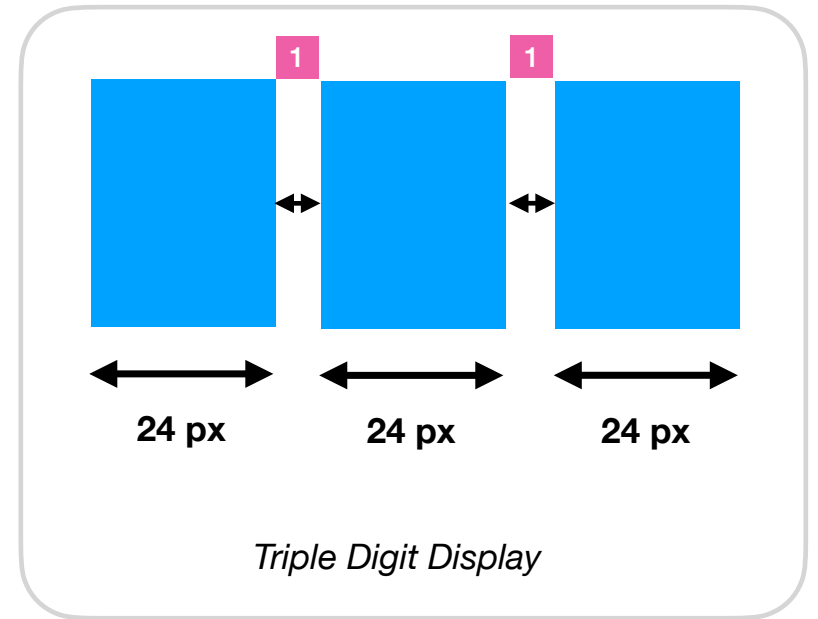
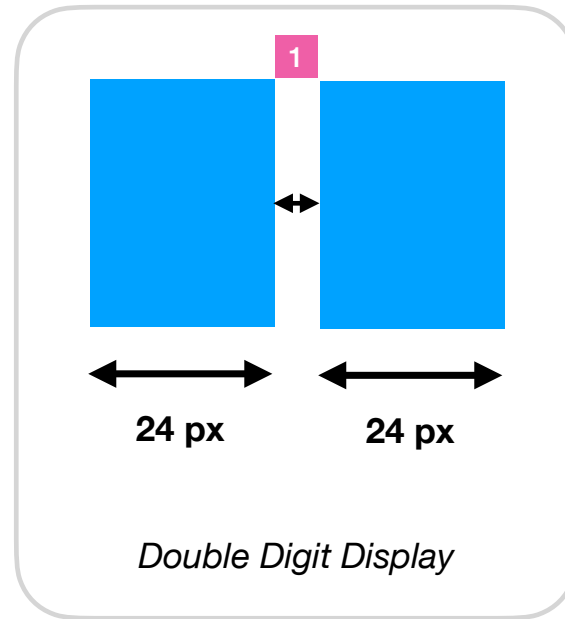
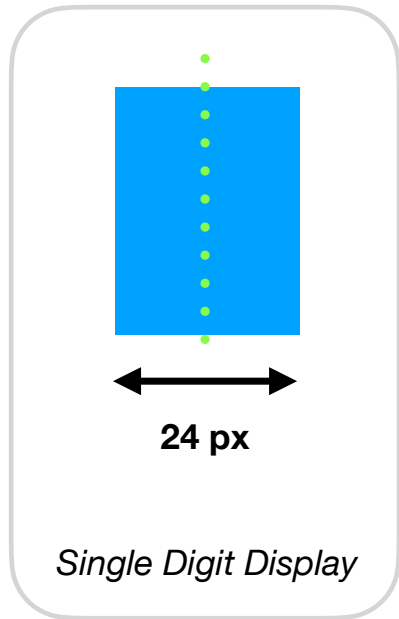


new Sprite(xpos, ypos)

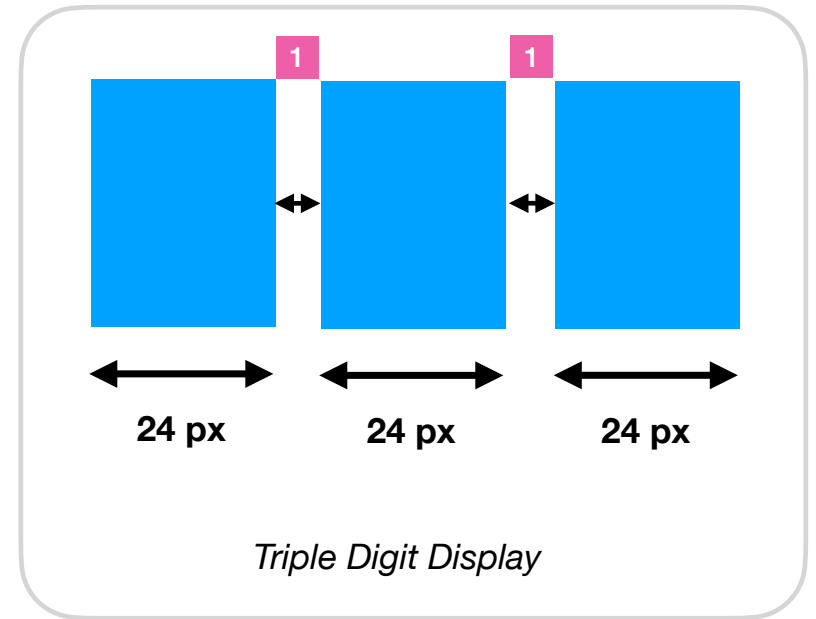
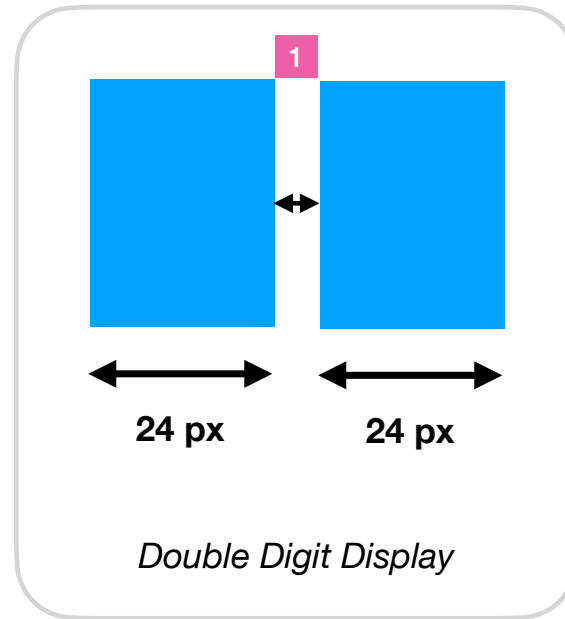
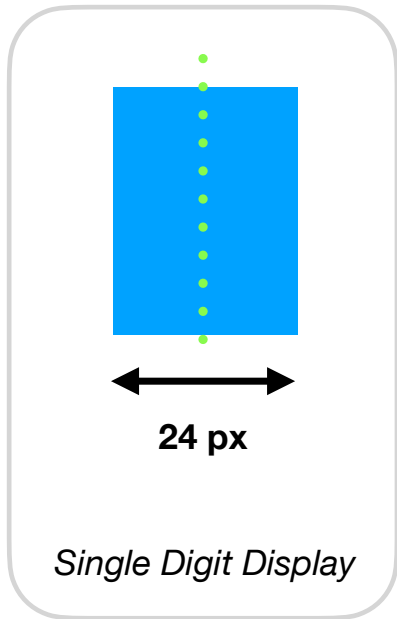
*xpos and ypos is in
the centre of the
given width and height.

Digit Image width: 24px

1 Leave a gap of 1 px between the digits

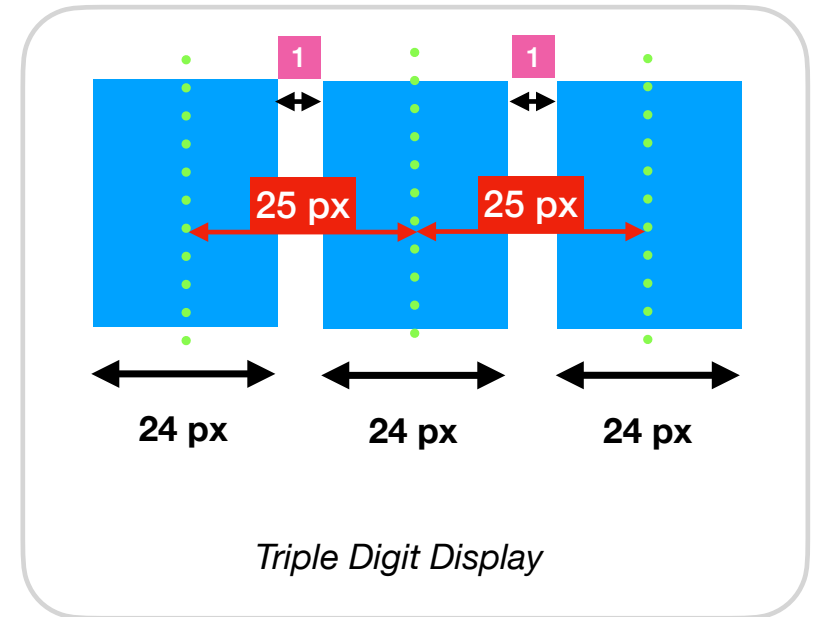
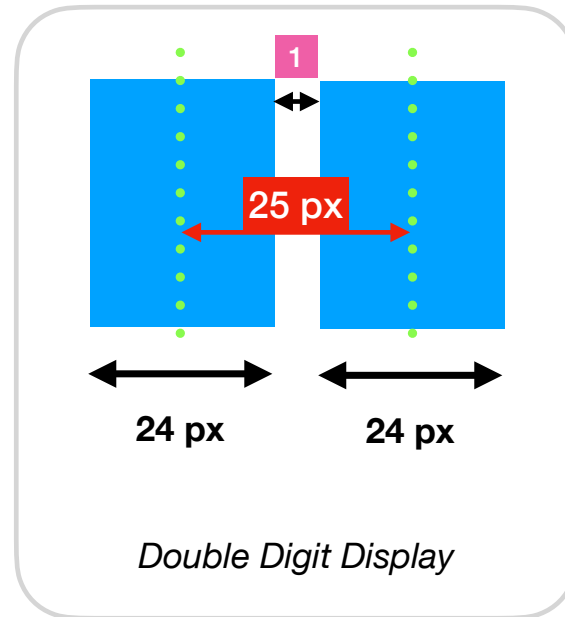
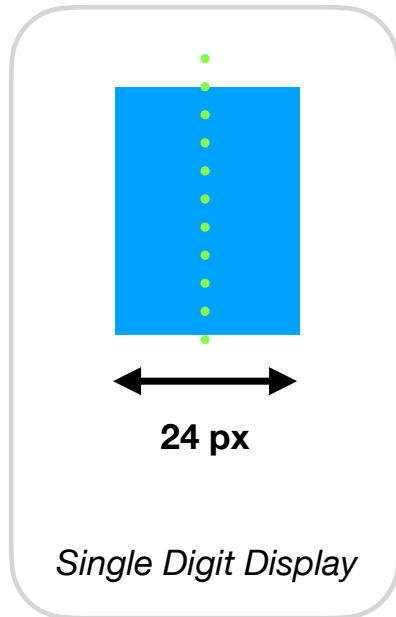


1 Leave a gap of 1 px between the digits

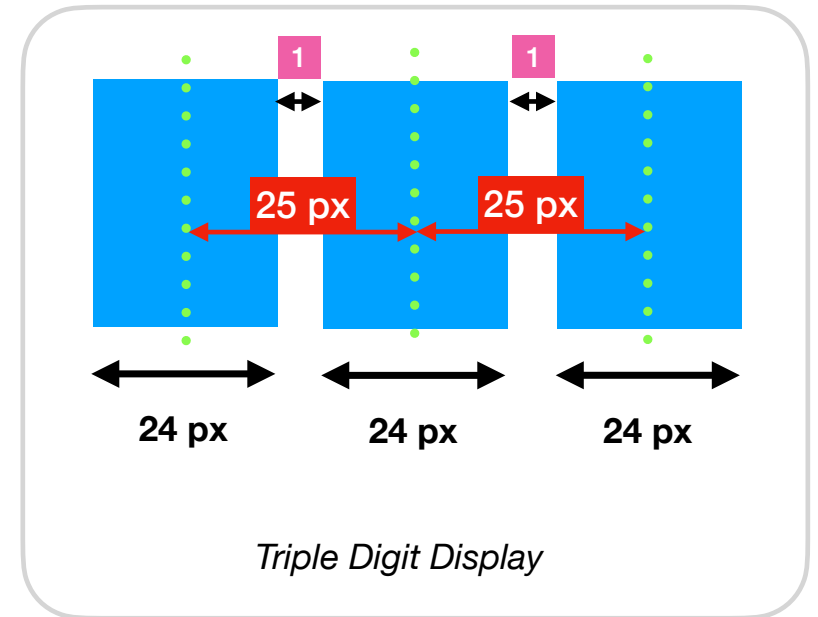
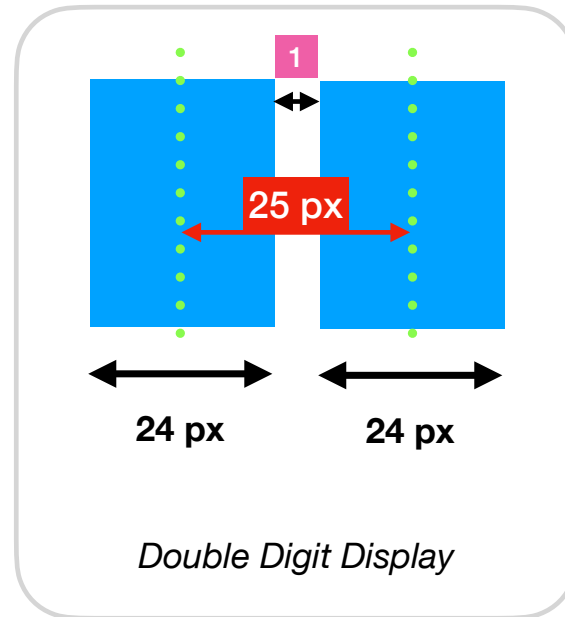
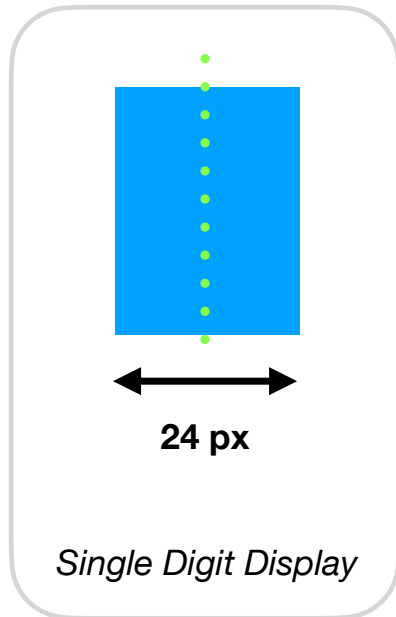


1. We can use Sprite Group, to manage all the digits in the Score value.

2. Considering the difference of 25 px, we can know each sprite's xpos (centre point)



2. Considering the difference of 25 px, we can know each sprite's xpos (centre point)

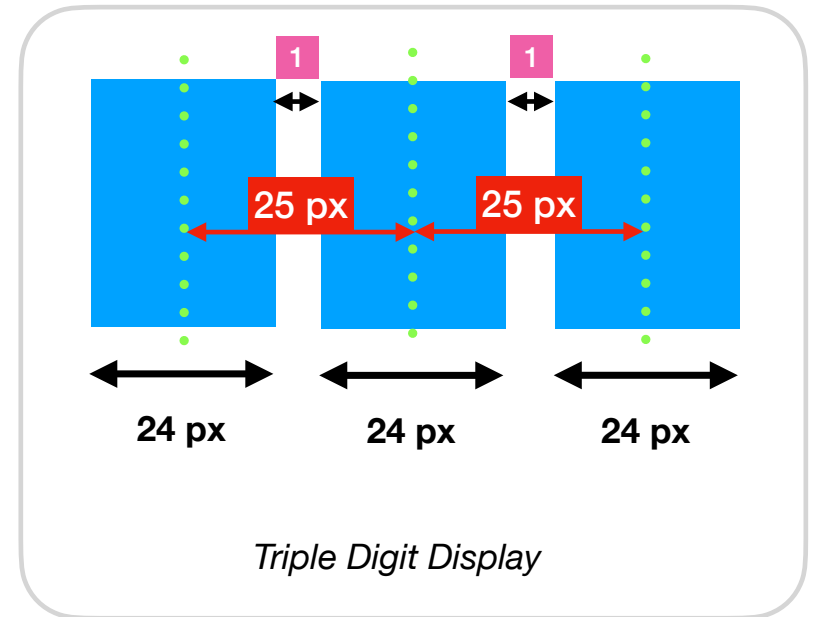
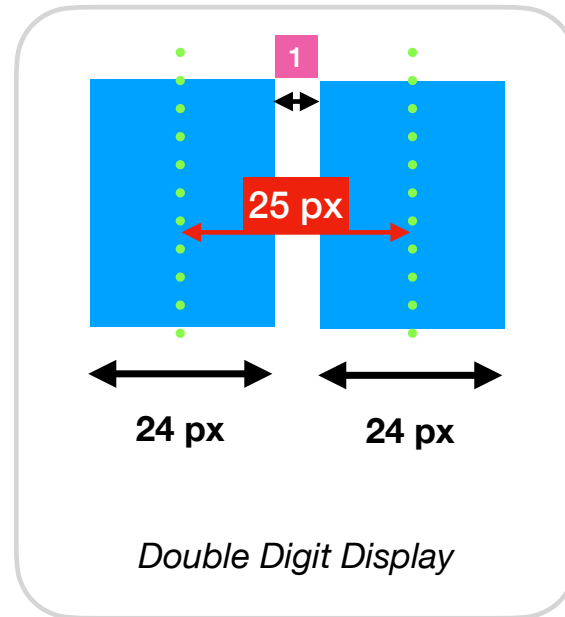
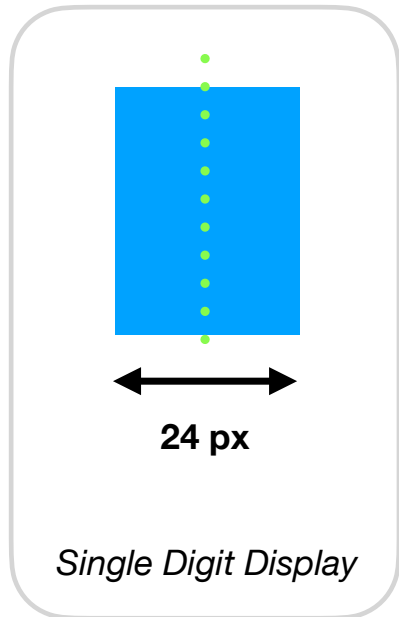


The pattern is...

1st Digit: 0×25

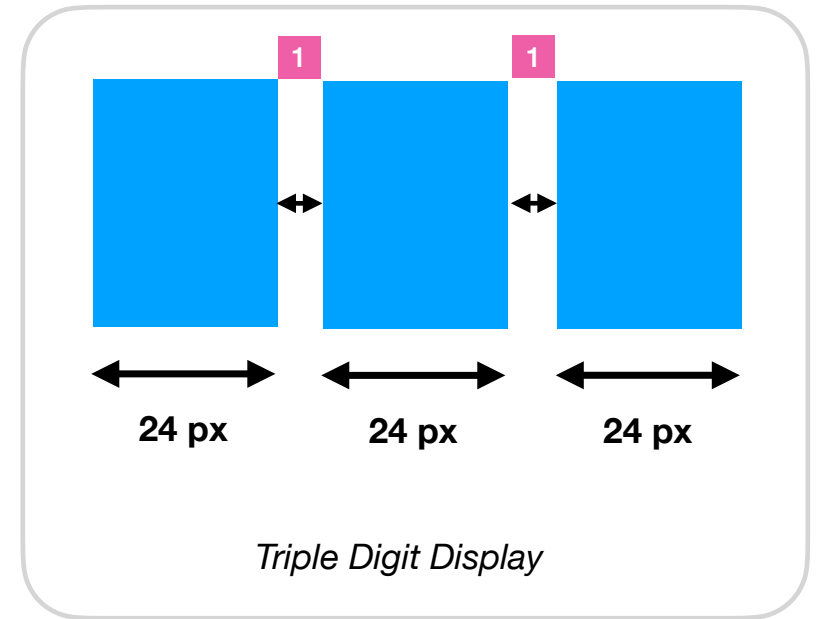
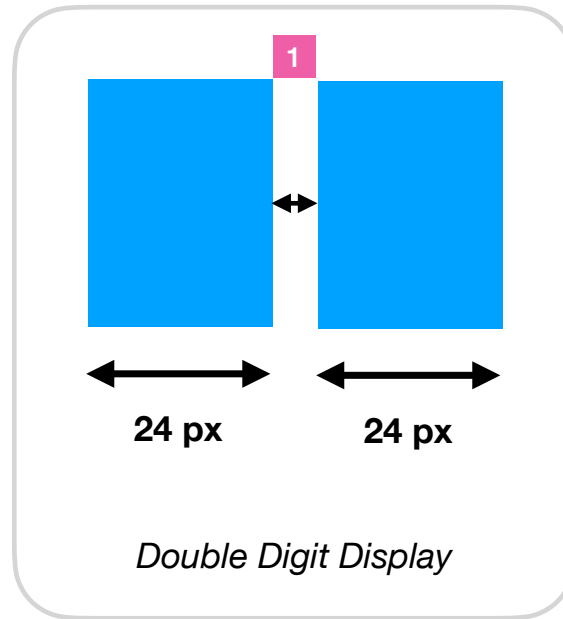
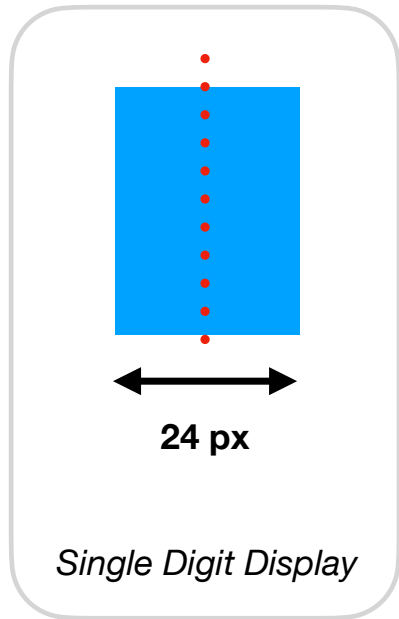
2nd Digit: 1×25

3rd Digit: 2×25



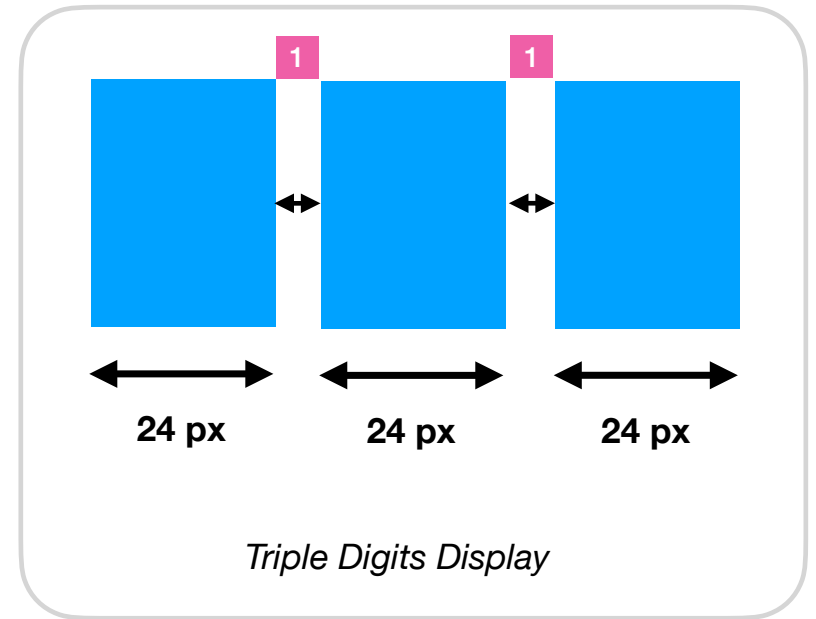
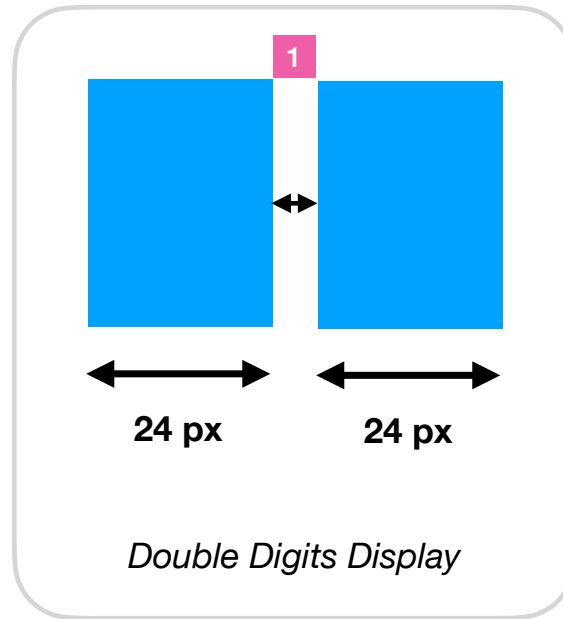
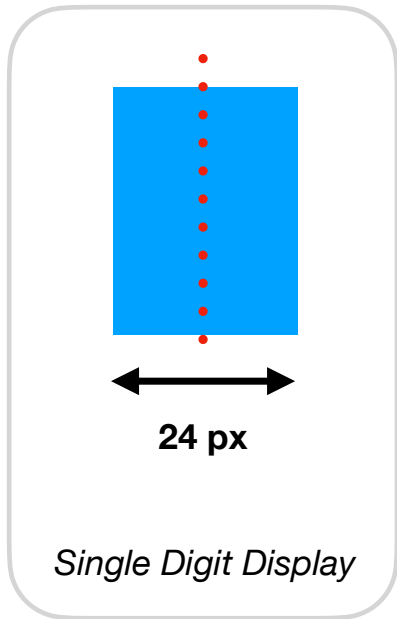
Now we know how to set xpos for each Sprite.

To find the middle value for 3 different test cases...



Single digit is always easy.

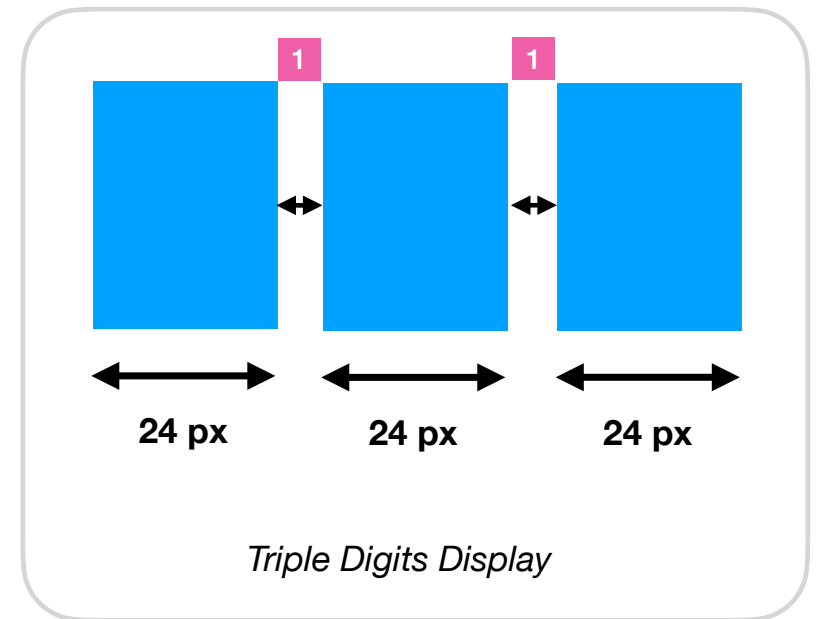
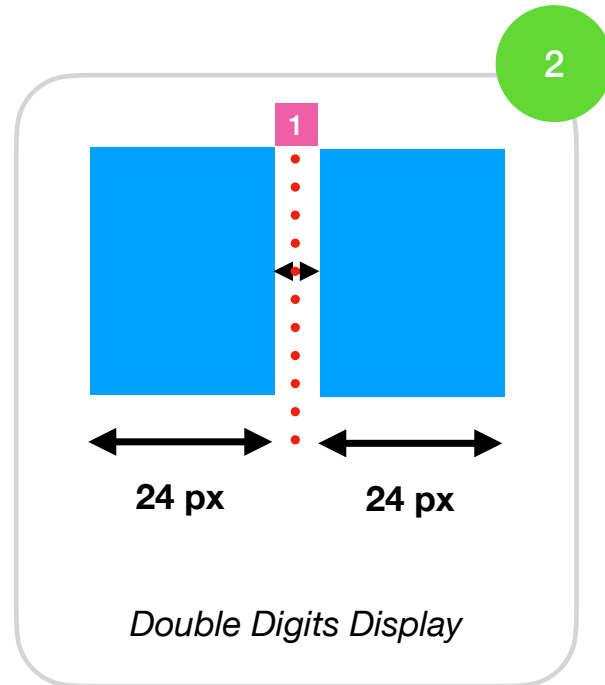
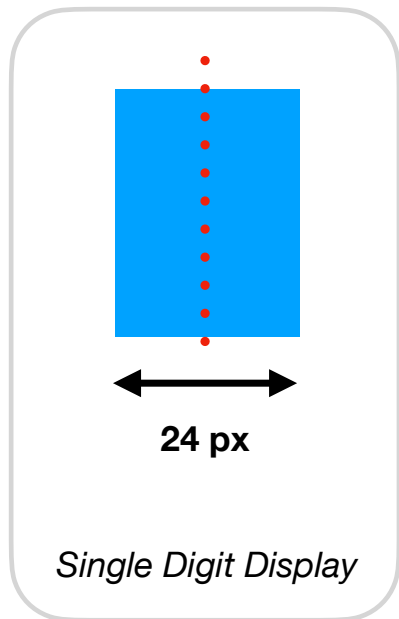
To find the middle value for 3 different test cases...



What about the middle point of each group?



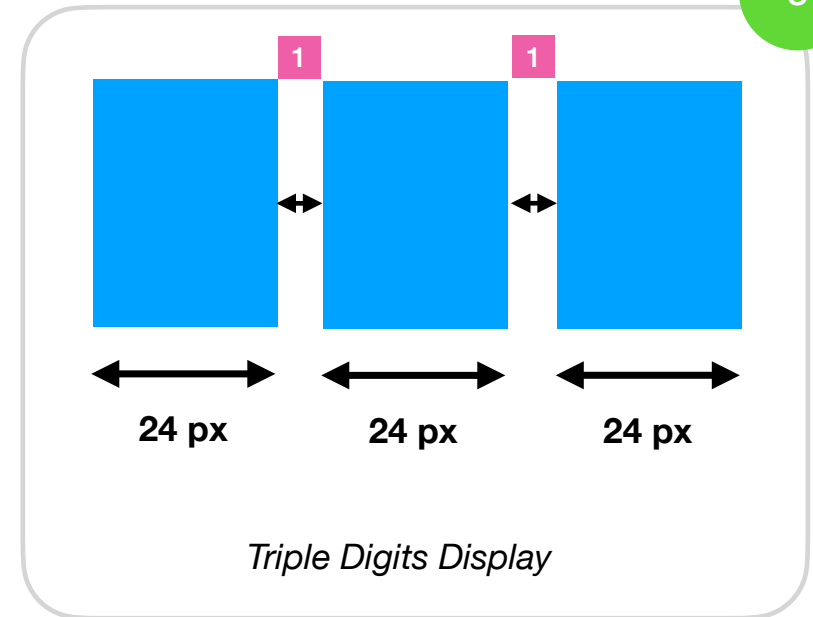
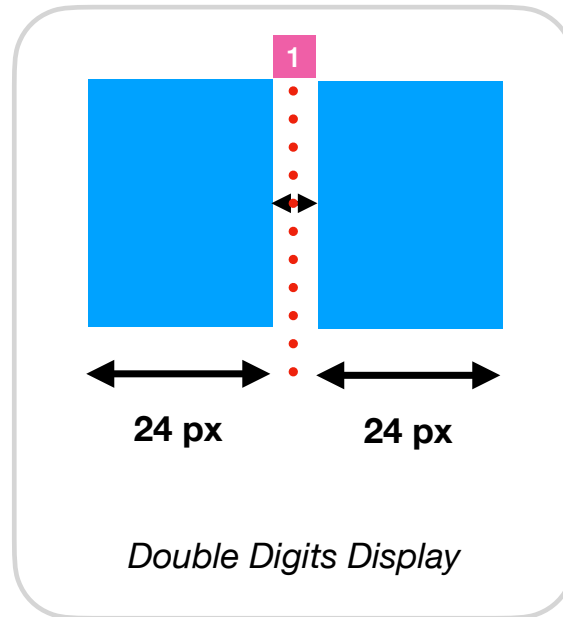
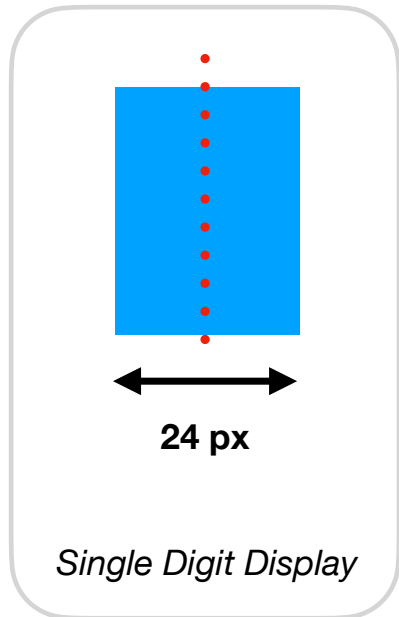
To find the middle value for 3 different test cases...



What is the middle point of this group?



To find the middle value for 3 different test cases...

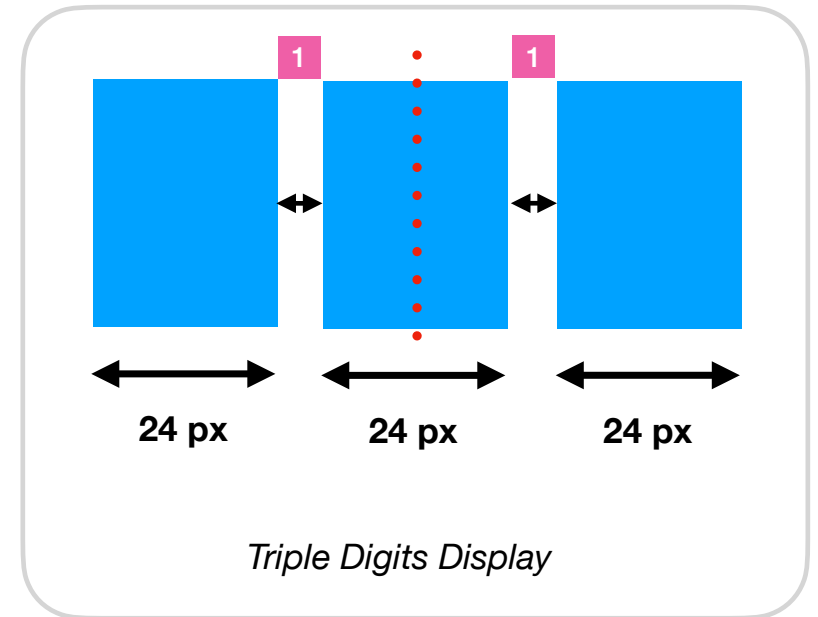
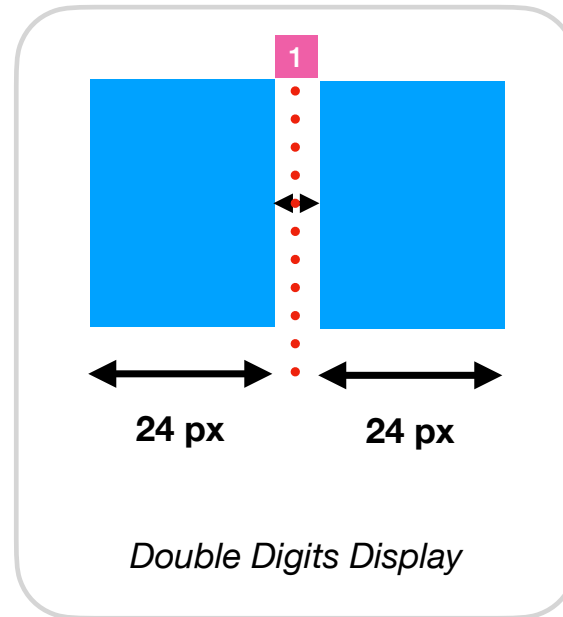
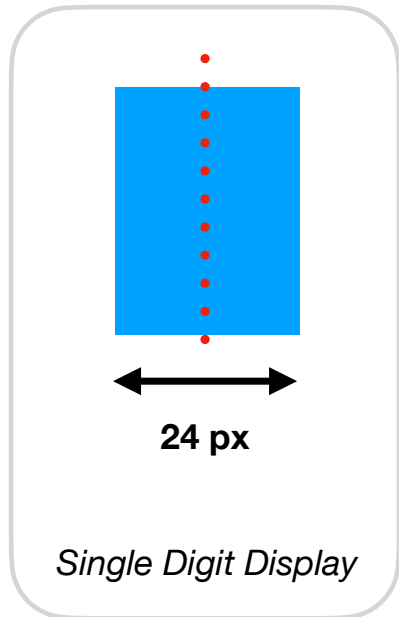


3

What is the middle point of this group?



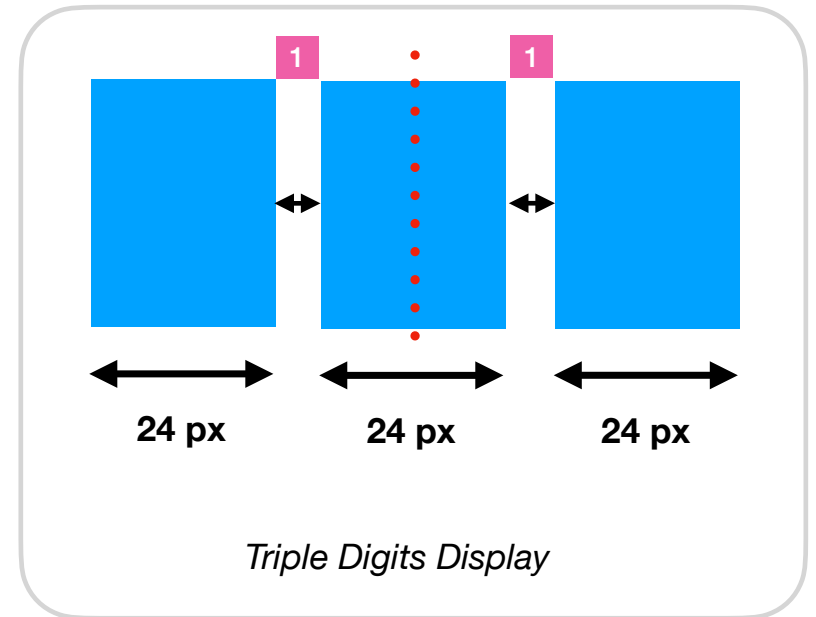
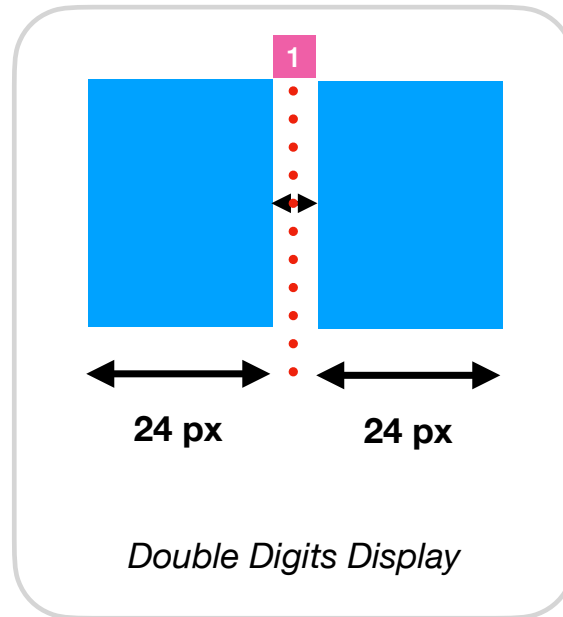
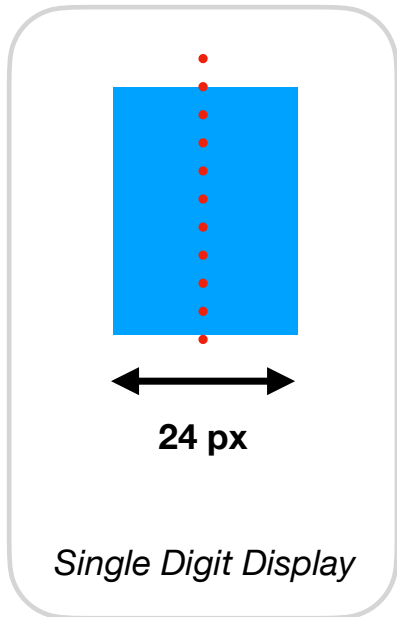
To find the middle value for 3 different test cases...



What is the middle point of this group?

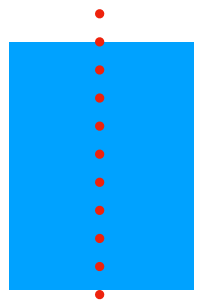


To find the middle value for 3 different test cases...



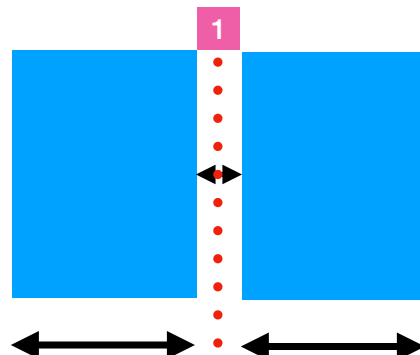
Did you see a pattern
in here?





24 px

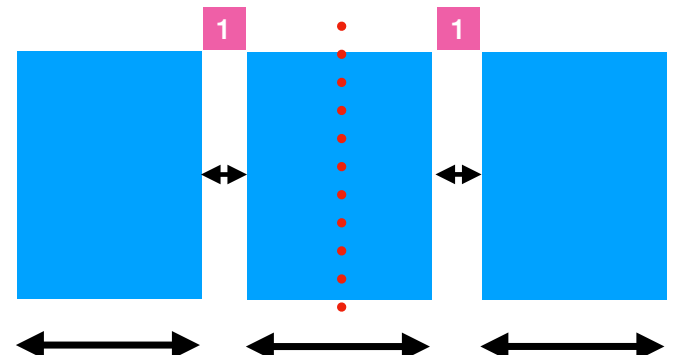
Single Digit Display



24 px

24 px

Double Digits Display



24 px

24 px

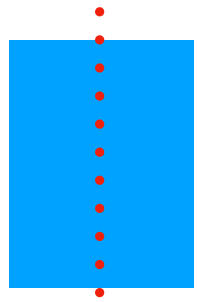
24 px

Triple Digits Display

Did you see a pattern
in here?

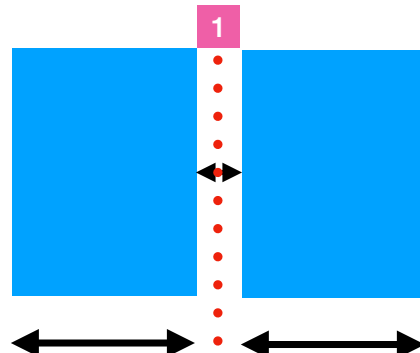


Find the total width of everything and divide by 2



24 px

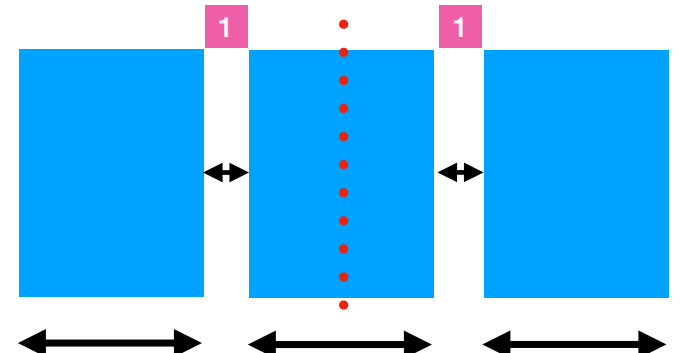
Single Digit Display



24 px

24 px

Double Digits Display



24 px

24 px

24 px

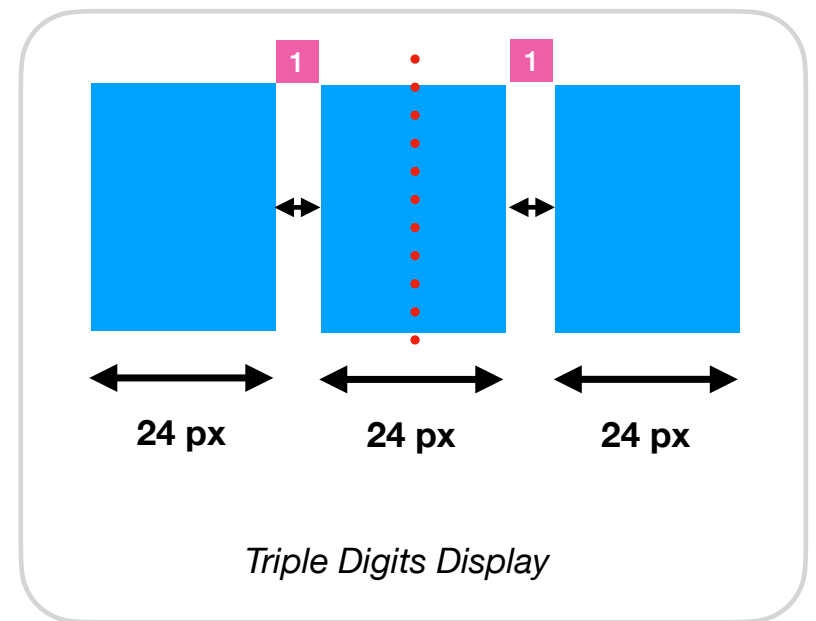
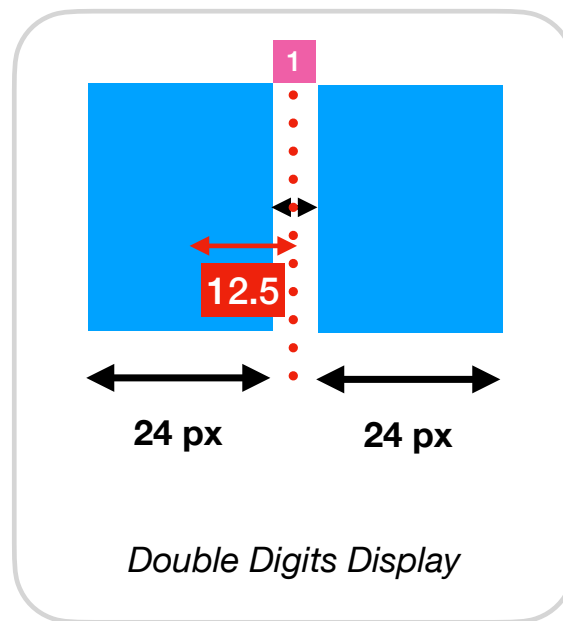
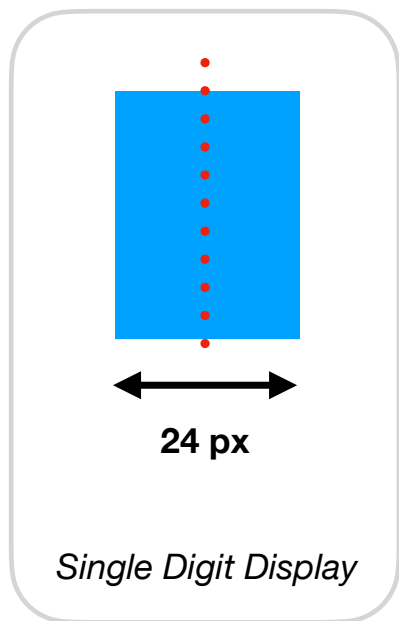
Triple Digits Display

Find the total width of everything and divide by 2

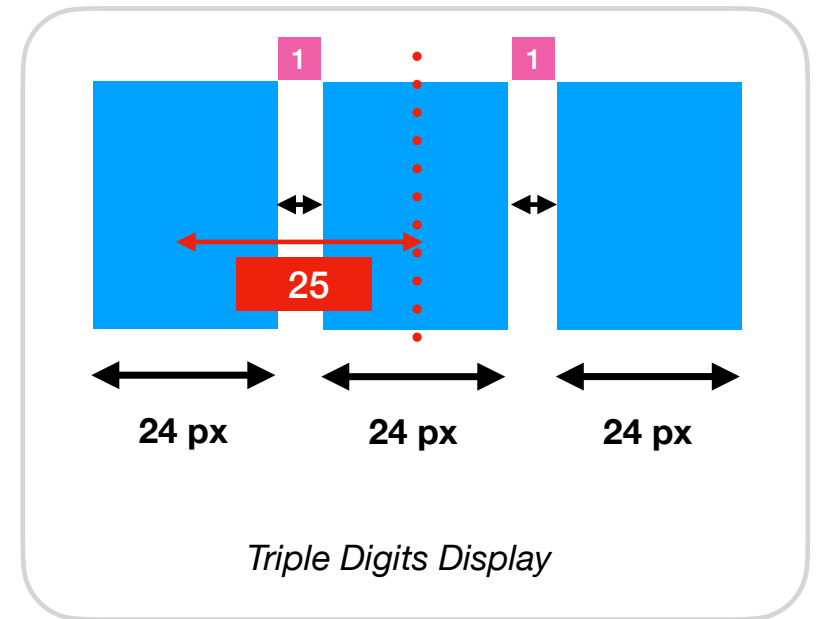
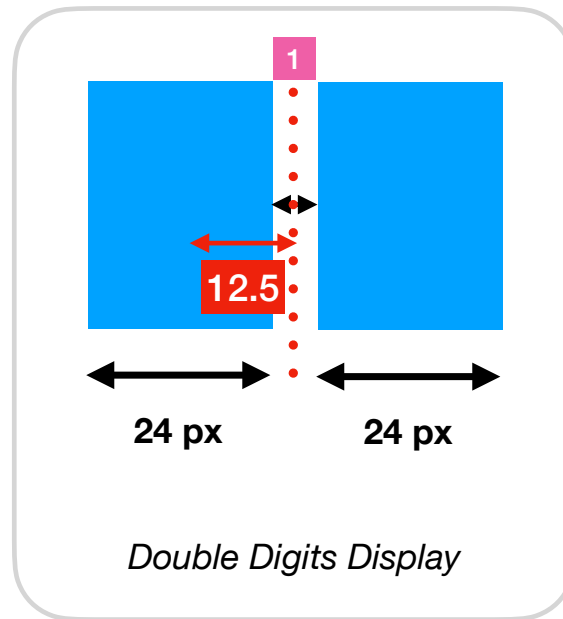
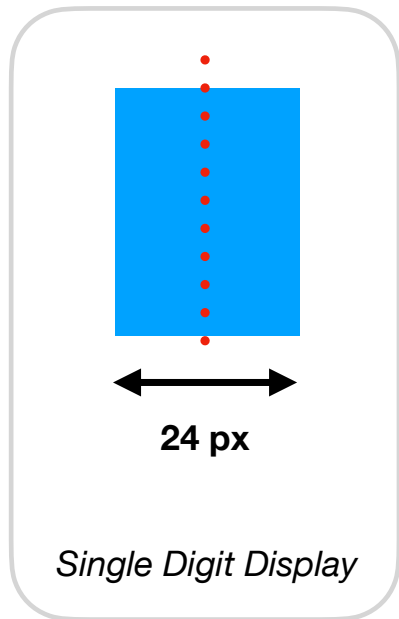
But we will need to
move each sprite to
new xpos one by one



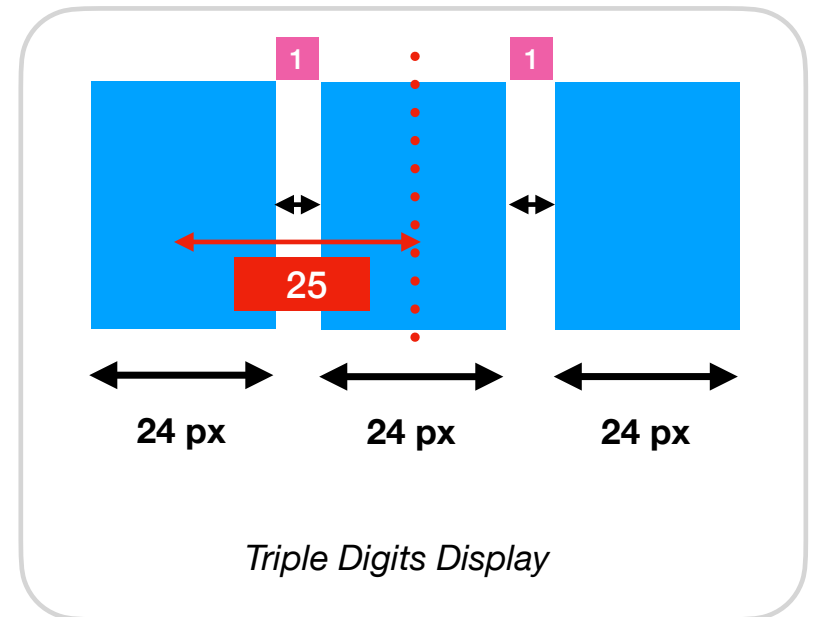
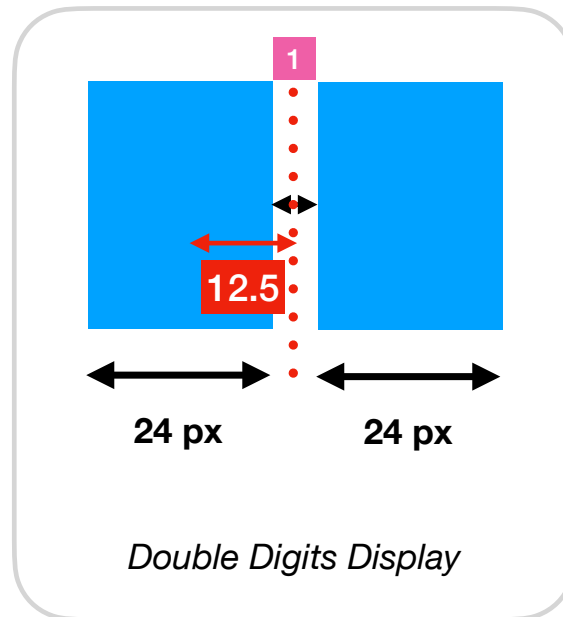
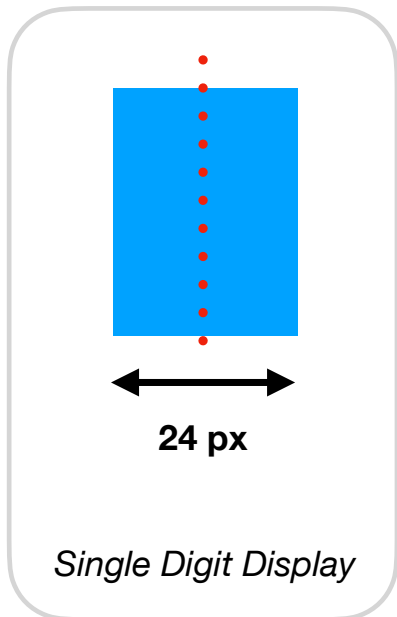
Move each sprite's X position one by one...



Move each sprite's X position one by one...



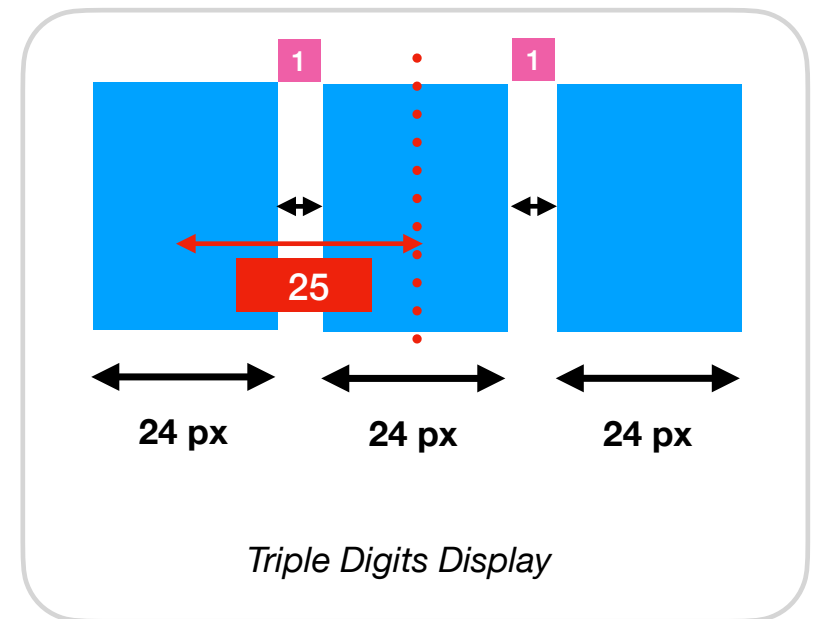
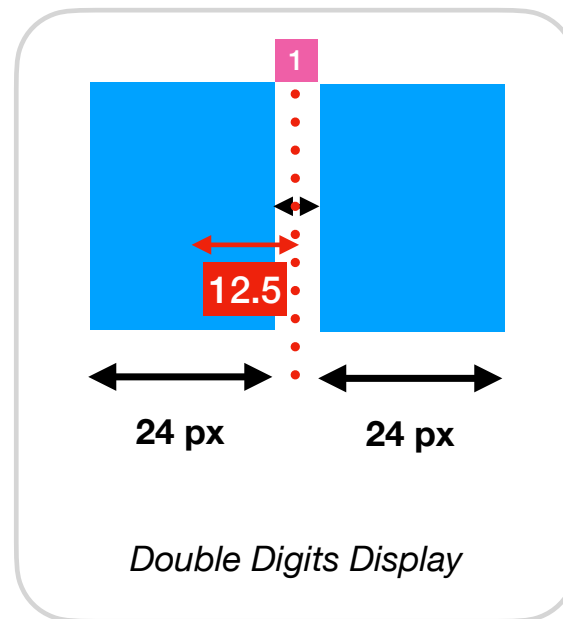
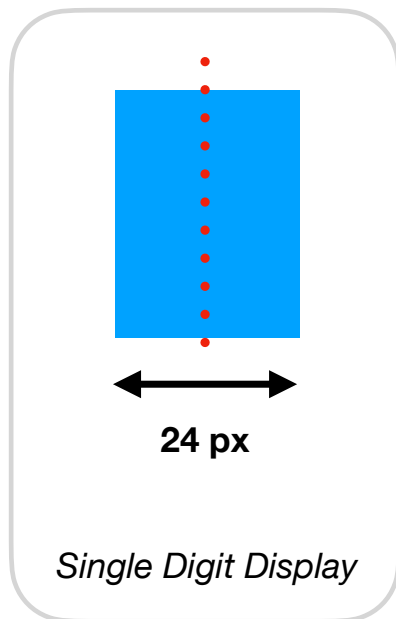
Move each sprite's X position one by one...



Did you see a pattern
in here?



Move each sprite's X position one by one...



$(\text{Number of digits} - 1) * 25 / 2$

Did you see a pattern
in here?



PREVIOUSLY we have...

Step 7: Use Sprite Group (Part 2)

```
59. // use function to group codes within a context together
```

```
60. function drawScore() {
```

```
62.     // clear previous score digits sprites
```

```
63.     scoreGroup.removeAll();
```



Remove all members

```
65.     // number to a String value
```

```
66.     let scoreString = str(scoreNumber);
```

```
67.     // String to an Array
```

```
68.     let scoreDigitArray = scoreString.split("");
```

```
70.     let offset = 0;
```

```
71.     let middle = width/2; // center of screen
```

```
73.     for (let one of scoreDigitArray) {
```

```
74.         let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);
```

```
75.         oneSprite.img = digitImgs[one];
```

```
76.         offset = offset + 25; // move right by size of image width + 1px
```

```
77.     }
```

```
78. }
```

```
79.
```

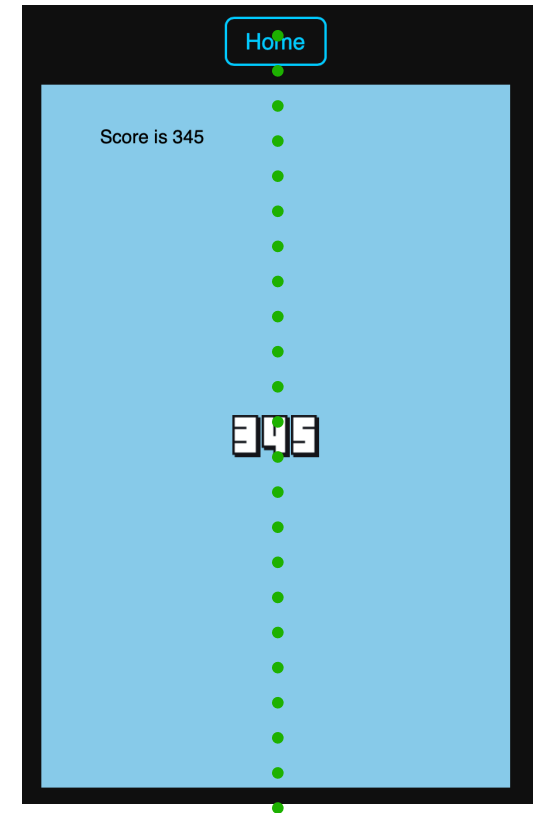


Add member to the Group

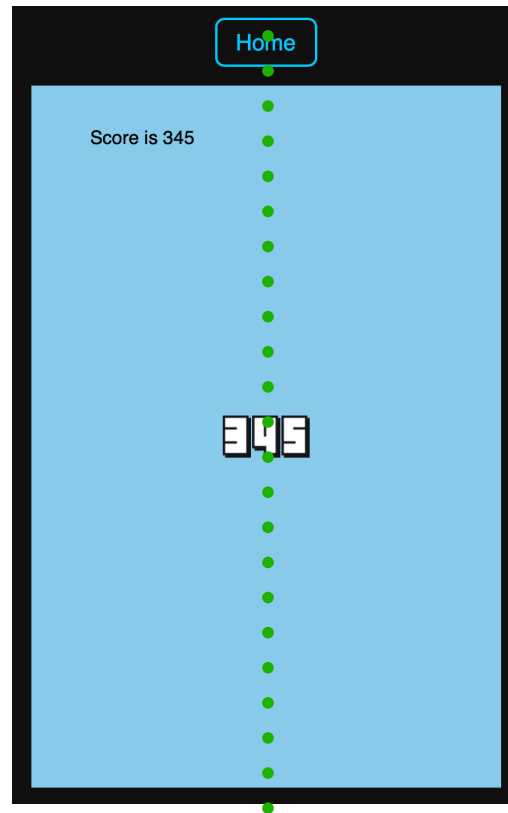
Now the codes looking like this...

```
73. for (let one of scoreDigitArray) {  
74.     let oneSprite = new scoreGroup.Sprite(middle+offset, height/2, 24, 36);  
75.     // move to center  
76.     oneSprite.x = oneSprite.x - (scoreDigitArray.length-1)*25/2;  
  
78.     oneSprite.img = digitImgs[one];  
79.     offset = offset + 25; // move right by size of image width + 1px  
80. }  
81.
```

Subtract using the formula



Observe: The digits are in the **centre** of the screen.



How to follow the camera?